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A.6 Deductibles & Limits

(1) Deductibles

- Let's say X is the amt. of loss
  Now, if there is a deductible of d,
  then the resulting payment is

$$\boxed{\text{Payment} = (X-d)_+ := \begin{cases} 0 & X \leq d \\ X-d & X > d \end{cases}}$$

- The uncovered cost to the insured is

$$\text{Uncovered cost} = \min(X, d) = X \wedge d$$

$$= \begin{cases} X & X \leq d \\ d & X > d \end{cases}$$

Total Loss = Insurance payment + Uncovered loss to insured

$$\boxed{X = (X-d)_+ + (X \wedge d)}$$

Eg. Loss is uniform on {1, 2, 3, 4, 5} & we have a deductible of 2. What E[Pay]? What is the probab. that UL will be 2.

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x	$P(X = x)$	Payment	UL
1		0	1
2		0	2
3	$1/5$	$3-2 = 1$	2
4		$4-2 = 2$	2
5		$5-2 = 3$	2

$X \geq d$, $\min(X, d) = d$

$$E[Pay] = 0 + 0 + \frac{1}{5} + \frac{2}{5} + \frac{3}{5} = \frac{6}{5} = 1.2$$

$$P[UL = 2] = \frac{4}{5}$$

- Sometimes, it is easier to find the payment than the uncovered loss.

$$\underset{\text{loss}}{X} = \underset{\text{pay}}{(X-d)_+} + \underset{\text{uncovered loss}}{(X \wedge d)}$$

$$\boxed{E[X] = E[(X-d)_+] + E[X \wedge d]}$$

But this concept only works for the first moment.

Eg. each week has 0 or 1 tornado. $P(T) = 0.3$, policy pays \$100/T. $d = \$50$. 8 weeks long tornado, they are independent. $E[Pay]$

$X = 100\underset{\text{no. of storms}}{N}$ $\qquad$ UL = 0 or 50

$$E[X] = 100N = 100\, E[N] \xrightarrow{} n \times p$$

$$= 100 \times 8 \times 0.3 = 240$$

$$E[X \wedge d] = 0 \cdot P(N=0) + 50 \cdot P(N \geq 1)$$

$$= 0 + 50 \cdot (1 - 0.7^8)$$

$$= 47.12$$

$$E[(X-d)_+] = 240 - 47.12 = 192.88$$

Eg. Loss N is geometric on $\{0, 1 \ldots\}$ with mean 2. Losses are insured for \$100. Annual deductible of \$150. $E[Pay]$

$$E[X] = 2 = \frac{1-p}{p} \Rightarrow p = 1/3$$

$$E[UL] = \cancel{0 \cdot P(N=0)} + 100 \cdot P(N=1) + P(N \geq 2) \cdot 150$$

$$= 100 \times (1-p)\, p + 150 \times (1-p)^2$$

$$p \overset{!}{=} 1/3$$

$$E[UL] = 88.9$$

$$E[N] = 100\, E[X] = 200$$

$$E[Pay] = 200 - 88.9 = 111.1$$

L2. <u>Limits</u>

- another way for the payment to be less than the total loss is to have a policy limit.

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$X \rightarrow$ loss amount, $u \rightarrow$ policy limit, if there is no deductible.

$$\text{Payment} = \begin{cases} X & X \leq u \\ u & u < X \end{cases}$$

in this case, $\boxed{\text{Payment} = \min\{X, u\} = (X \wedge u)}$

- If both deductible d & limit u is present,

$$\text{Payment} = \begin{cases} 0 & X \leq d \\ X-d & d < X \leq d+u \\ u & d+u \leq X \end{cases}$$

- The expected payment is c/d the net premium or the benefit premium.

Eg. Losses ~ Poisson (2.4). Loss gives 50 in damage. $u = 75$. $E[\text{Pay}] = ?$

$N \rightarrow$ no. of loss

$N = 0$ $\quad$ Pay $= 0$

$N = 1$ $\quad$ Pay $= 50$

$N \geq 2$ $\quad$ Pay $= 75$

$$E[\text{Pay}] = 0 + 50 \cdot P(N=1) + 75 \cdot P(N \geq 2)$$
$$= 50 \times 2.4e^{-2.4} + 75(1 - e^{-2.4} - 2.4e^{-2.4})$$
$$= 62.75$$

Eg. $X \sim \text{Bino.}(5, 0.4)$, $d = 1$, limit $= 3$

$E[\text{Pay}]$ of any loss

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x	0	1	2	3	4	5
$P(X=x)$	$(0.6)^5$	$5\times0.4\times(0.6)^4$	- - -			
Pay	0	0	1	2	3	3
UL	0	1	1	1	1	2

$E[Pay] = 1.06752$ $\qquad$ $E[UL] = 0.93248$

$$E[Pay] = 5\times0.4 - E[UL] = 1.06752$$

- You can use TI-30XS calculator for calculating the diductibles.